

Genomska analiza

Šta je genomska analiza?

- Šta je genom?
 - Kompletan genetski kod jednog organizma



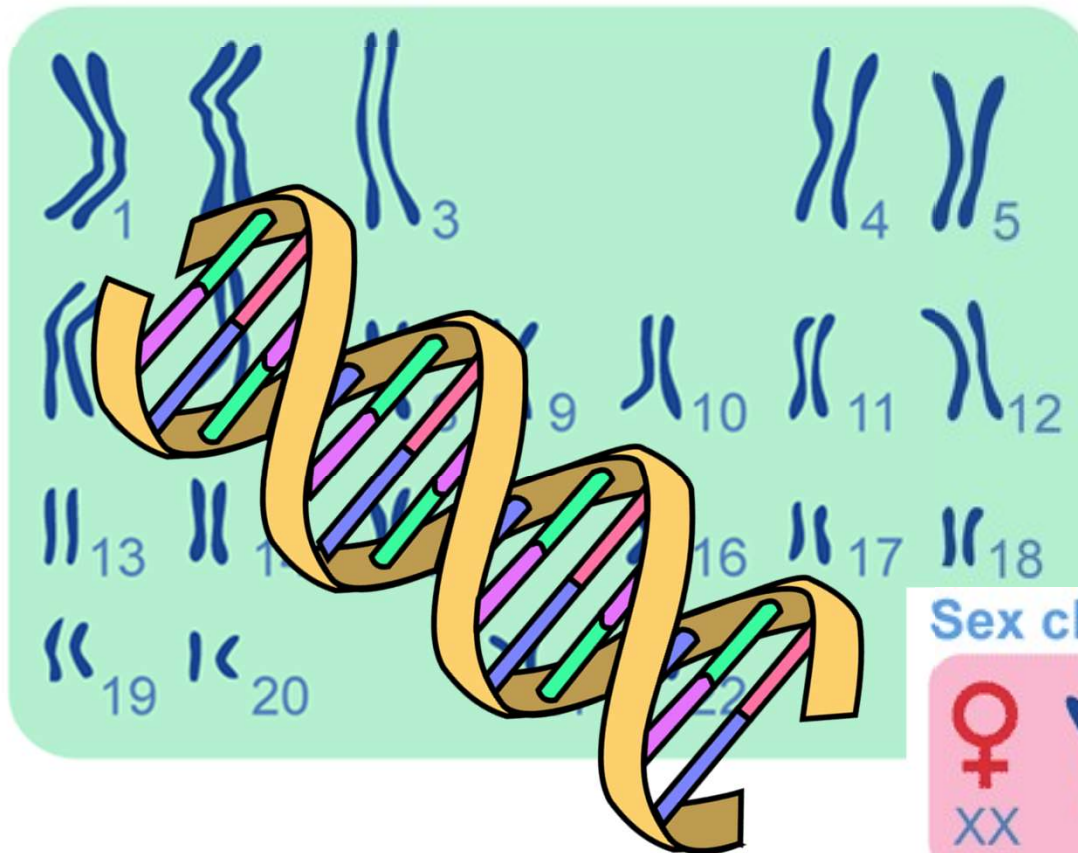
```
CCTCCTCAGTGCCACCCAGCCCACTGGCAGCTCCCAAACA  
GGCTCTTATTAACACCCCTGTTCCCTGCCCTTGGAGTG  
AGGTGTCAAGGACCTAACTAAAAAAGAAAA  
AGAAAAGAAAAAATTTAAATTTAAGTAATCTTTGAA  
AAAACTAATTTCTAAGCTTCTTCATGTCAAGGACCTAATG  
TGCTAACAGCACTTTTGTGACCATTTTGGATCTGAAA  
GAAATCAAGAATAATGAAGGACTTGATACATTGGAAGA  
GGAGAGTCAAGGACCTACAGAAAAAAGAAAA  
AAGAAAAGAAAAAGATTTAAATTTAAGTAATCTTTGA  
AAAACTAATTTCTAAGCTTCTTCATGTCAAGGACCTAAT  
GTCTGTGTTGCAGGTCTTCTGCATTTCCCTGTCAAAGA  
AAAAGAATTTAAATTTAAGTAATCTTTGAAAAAATA  
ATTTCTAAGCTTCTTCATGTCAAGGACCTAATGTCAGGCC  
GGCTCTTATTAACACCCCTGTTCCCTGCCCTTGGAGTG
```

- Koliko memorije zauzima?
 - ~3.2 milijarde baza



Ljudski hromozom

Autosomes



█ = Adenine

█ = Thymine

█ = Cytosine

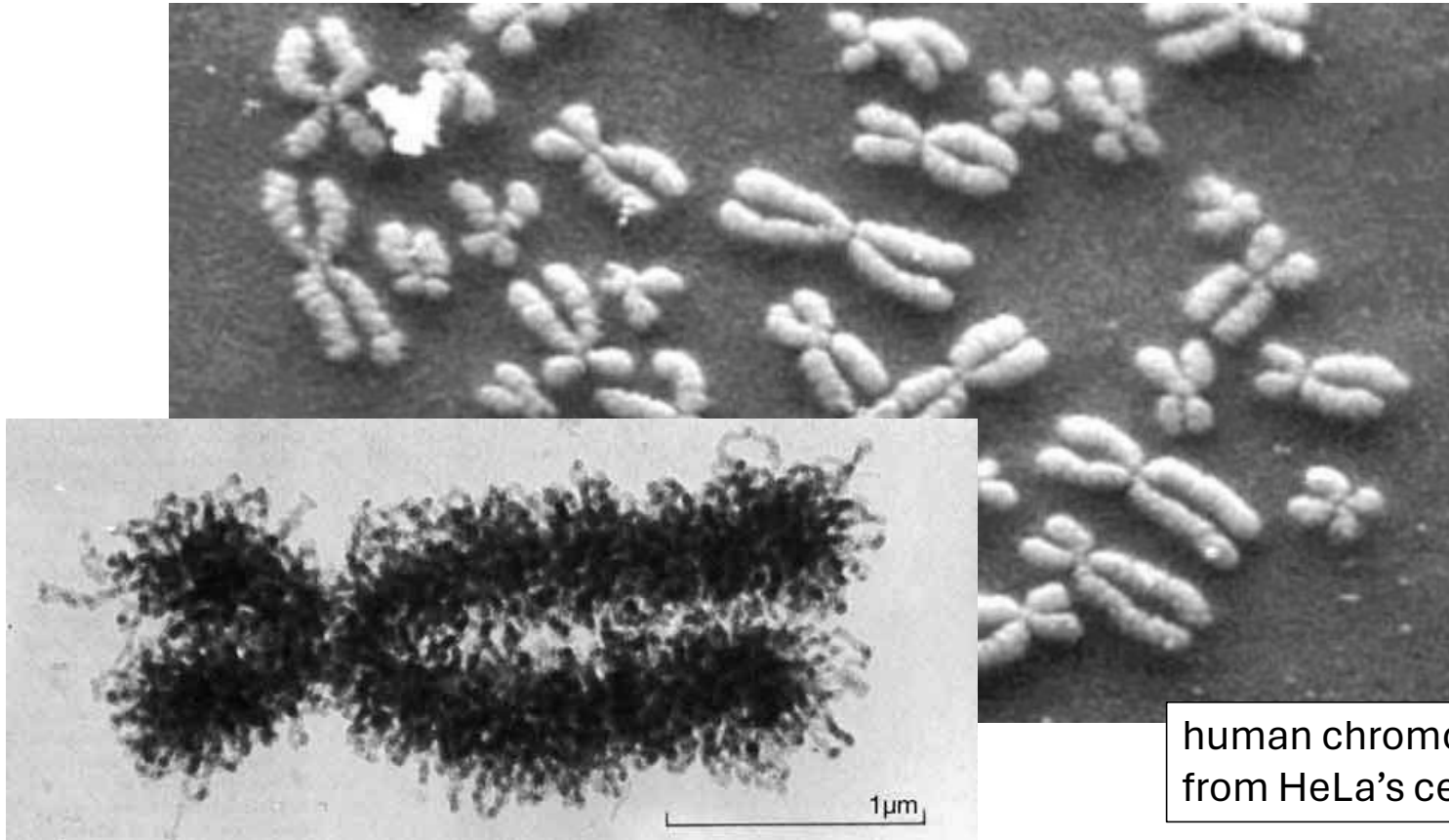
█ = Guanine

█ = Phosphate backbone

Sex chromosomes

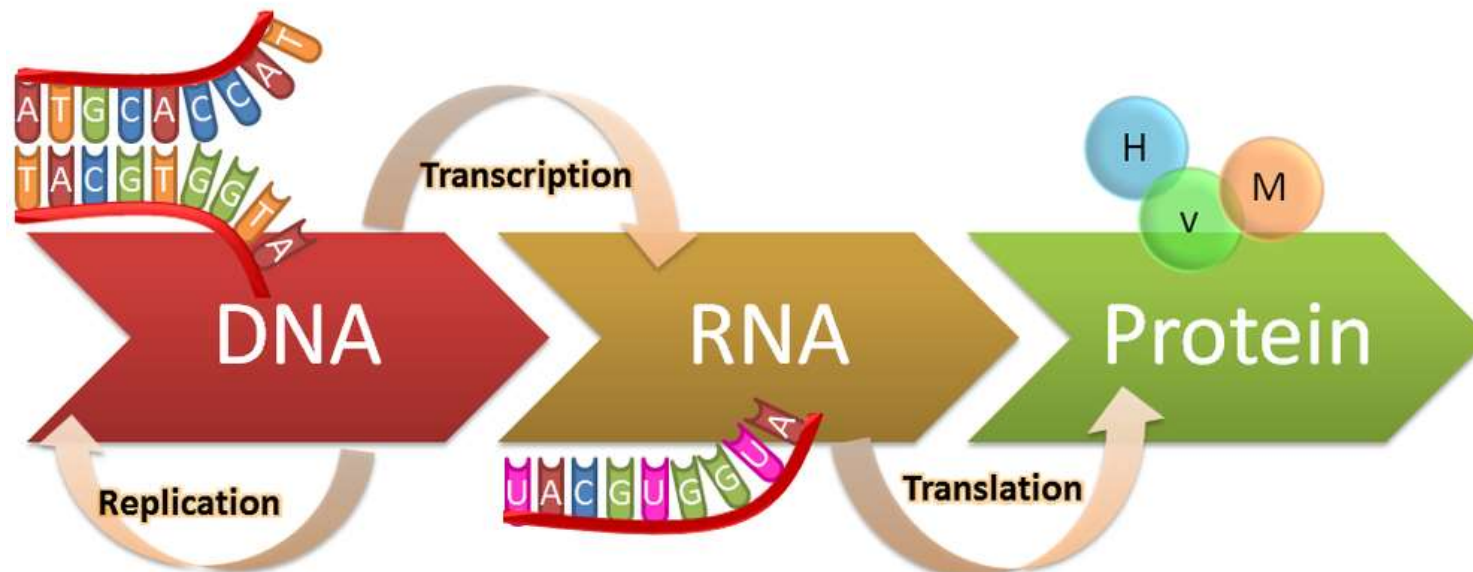


DNK pod mikroskopom



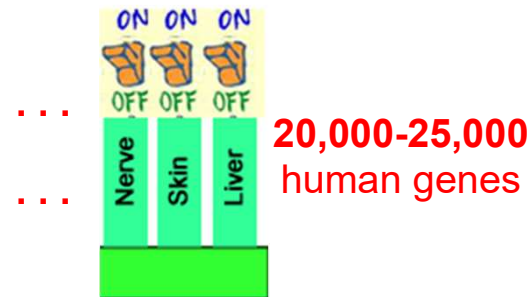
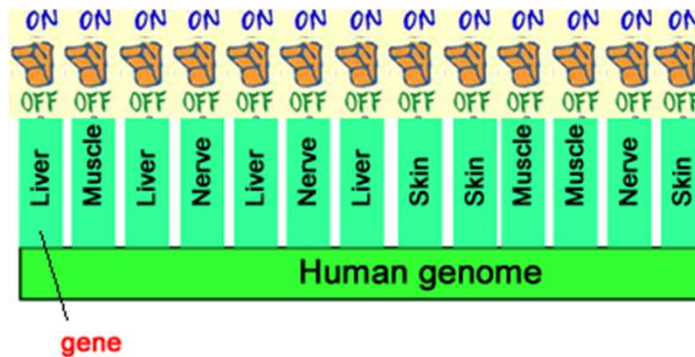
human chromosome #12
from HeLa's cell

Kako se koristi genetski materijal



Različiti organi i tkiva

- Sve ćelije imaju isti DNK i iste gene
 - Svaka ćelija koristi samo deo gena



NIH 2009 National DNA Day

Primene (1)

	SNP1	SNP2	Blood Pressure
Individual #1	...ACATG C CGACATTTCATAG G GCC...		180
Individual #2	...ACATG C CGACATTTCATAG A GCC...		175
Individual #3	...ACATG C CGACATTTCATAG G GCC...		170
Individual #4	...ACATG C CGACATTTCATAG A GCC...		165
Individual #5	...ACATG C CGACATTTCATAG G GCC...		160
Individual #6	...ACATG C CGACATTTCATAG G GCC...		145
Individual #7	...ACATG C CGACATTTCATAG A GCC...		140
Individual #8	...ACATG C CGACATTTCATAG A GCC...		130
Individual #9	...ACATG T CGACATTTCATAG G GCC...		120
Individual #10	...ACATG T CGACATTTCATAG A GCC...		120
Individual #11	...ACATG T CGACATTTCATAG G GCC...		115
Individual #12	...ACATG T CGACATTTCATAG A GCC...		110
Individual #13	...ACATG T CGACATTTCATAG G GCC...		110
Individual #14	...ACATG T CGACATTTCATAG A GCC...		110
Individual #15	...ACATG T CGACATTTCATAG G GCC...		105
Individual #16	...ACATG T CGACATTTCATAG A GCC...		100



SNP: single nucleotide polymorphism

Primene (2)

- Strukturalne varijacije



OBESITY

Walters, *Nature* 2010
Deletion of 593 kb



Deletion in the short arm
of chromosome 16 (16p11.2)



UNDERWEIGHT

Jacquemont, *Nature* 2011
Duplication of 593 kb



Duplication in the short arm
of chromosome 16 (16p11.2)

Primene(3)

- Razumevanje genetskih variacija
- Nadgledanja širenja bolesi
- Personalizovana medicina
- Arhitekture za smeštanje podataka
- ...

Šta zahteva genomska analiza?

- Brzina/propusnost
 - Analiza u realnom vremenu
- Genomska analiza na nivou populacije
 - Skalabilnost
- Prenosni uređaj i energetska efikasnost
 - Specijalizovan hardver
- Privatnost
 - Iz DNK se mogu izvući mnoge informacije
- Preciznost
 - Greške svesti na minimum

Kako se radi analiza? [1]



CCCCCTATATATACGTACTAGTACGT
ACGACTTTAGTACGTACGT
TATATATACGTACTAGTACGT
ACGTACGCCCCTACGTA
TATATATACGTACTAGTACGT
ACGACTTTAGTACGTACGT
TATATATACGTACTAAAGTACGT
TATATATACGTACTAGTACGT
ACG TTTTTTAAACGTA
TATATATACGTACTAGTACGT
ACGACGGGGAGTACGTACGT



1×10^{12} bases



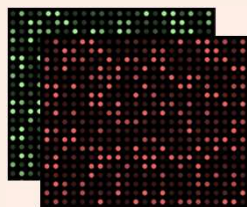
44 hours



<1000 \$

Različiti postupci

Illumina



Multiple images



.BCL/.CBCL

ONT

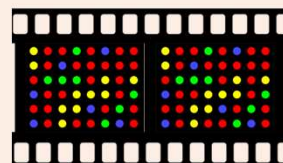


Squiggle



.FAST5

PacBio

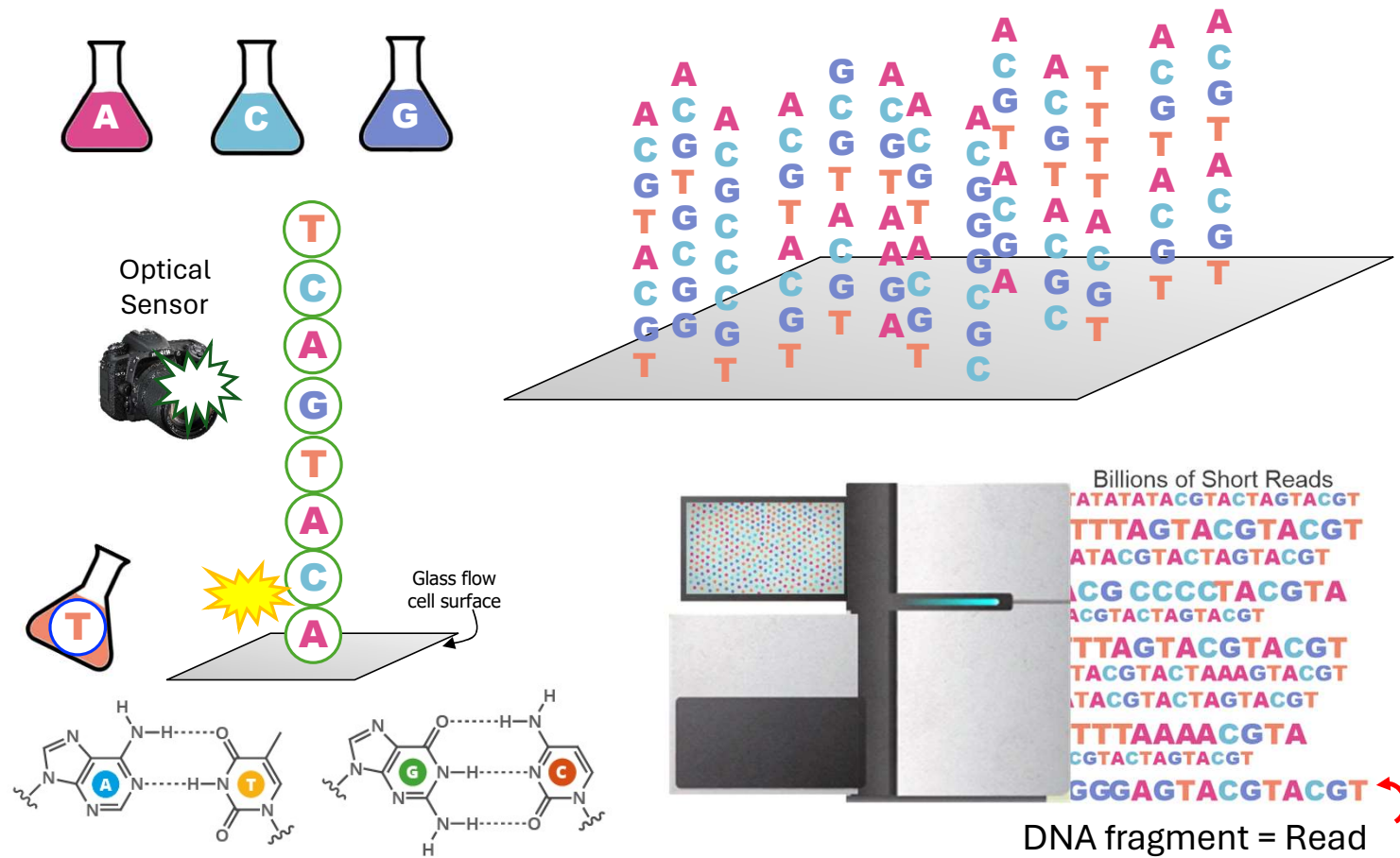


30-hour movie



.BAM

Illumina Machine



Sklapanje slagalice



Reference
genome



Reads

Tehnologija diktira algoritme

Short Reads (Illumina)

1 Sequencing Library preparation: 6.5 hours Sequencing: 68.2 Gb/hour	2 Basecalling 104.4 Gb/hour	3 Quality Control 1339.2 Gb/hour	4 Read Mapping 0.2 Gb/hour	5 Variant Calling 1.2 Gb/hour
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Ultra-long Reads (ONT)

1 Sequencing Library preparation: 24 hours Sequencing: 4.1 Gb/hour	2 Basecalling 0.833 Gb/hour	3 Quality Control 3420 Gb/hour	4 Read Mapping 1.7 Gb/hour	5 Variant Calling 0.044 Gb/hour
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Accurate Long Reads (PacBio)

1 Sequencing Library preparation: 24 hours Sequencing: 5.3 Gb/hour	2 Basecalling 8.3 Gb/hour	3 Quality Control 1081 Gb/hour	4 Read Mapping 1.4 Gb/hour	5 Variant Calling 1.1 Gb/hour
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Podaci

- Ljudski genom – referentni genom
 - https://www.ncbi.nlm.nih.gov/datasets/genome/GCF_000001405.40/
- Primeri izlaza aparata
 - <https://www.ncbi.nlm.nih.gov/sra/ERR240727>

Mapiranje pročitanih podataka

- Potpuna pretraga
 - $O(m^2kn)$ – m dužina jednog dela, k broj delova, n dužina referentnog genoma

Reference



Read

Efikasnija pretraga

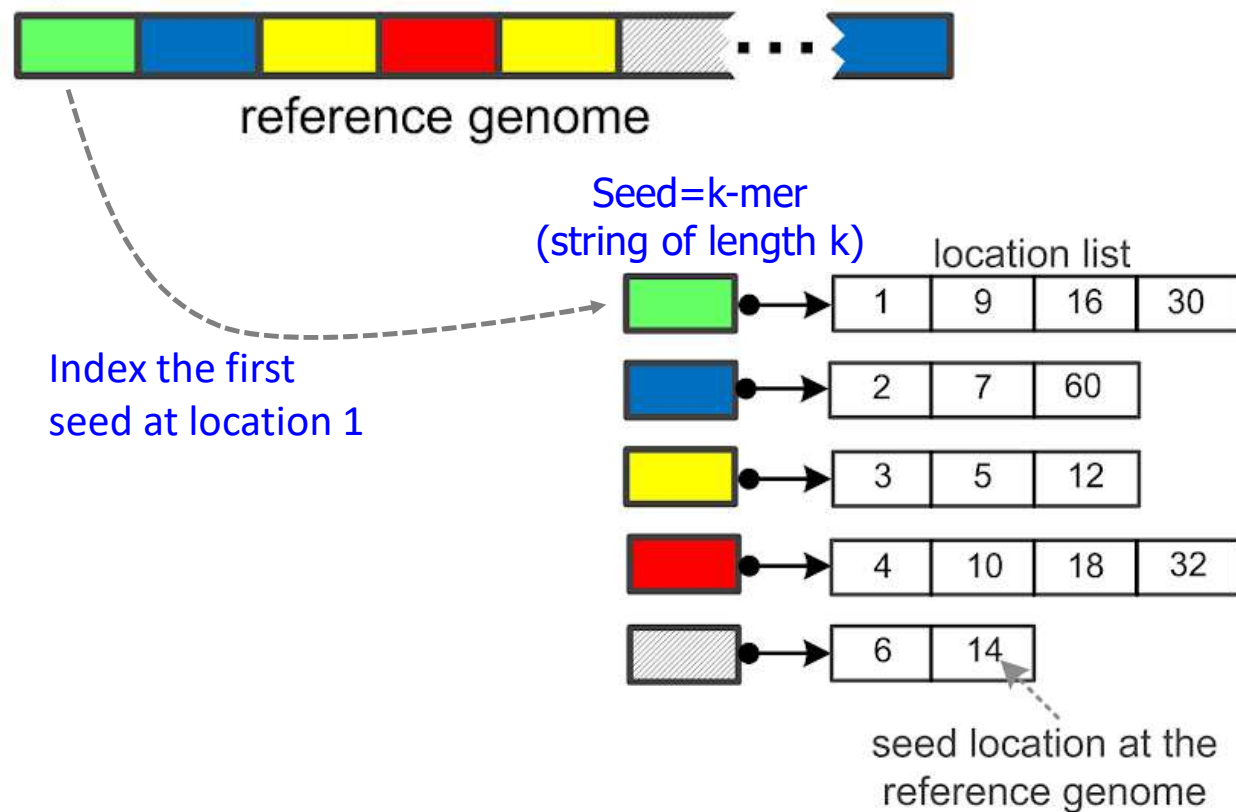
.FASTA file:

```
>NG_008679.1:5001-38170 Homo sapiens paired box 6 (PAX6)
ACCCCT[redacted]TCATTGACATTTAAACTCTGGGGCAGG[redacted]GAACGCGGCTGTCAGATCT
GCCACTTCCCCTGCCGAGCGGCGGTGAGAAGTGTGGGAACCGGCGCTGCCAGGCTCACCTGCCTCCCCGC
CCTCCGCTCCCAGGTAACCGCC[redacted]CCCCGGCCCGGCTCGGGGCCGCGGGGCTCTCCGCTG
CCAGCGACTGCTGTCCCCAAATCAAAGCCCGCCCCAAGTGGCCCCGGGGCTTGATTTTTGCTTTTAAAAG
GAGGCATACAAAGATGGAAGCGAGTTACTGAGGGAGGGATAGGAAGGGGGGTGGAGGAGGGACTTGTCTT
TGCCGAGTGT[redacted]CAAAGTAGCA[redacted]CTCCTA[redacted]TCCAGTCCGGCCCT
GAGCTGGGAGTAGGGGGCGGGAGTCTGCTGCTGTCTGCTAAAGCCACTCGCGACCGCGAAAAATGCA
GGAGGTGGGGACGCACTTTGCATCCAGACCTCCTCTGCATCGCAGTTC[redacted]CGCTTGGGAAAG
TCCGTACCCGCGCCT[redacted]AAAGACACCCTGCCGCGGTTCGGGCGAGGTGCAGCAGAAGTTTCCC
GCGGTTGCAAAGTGCAGATGGCTGGACCGCAACAAAGTCTAGAGATGGGGTTCGTTTCTCAGAAAGACGC
```

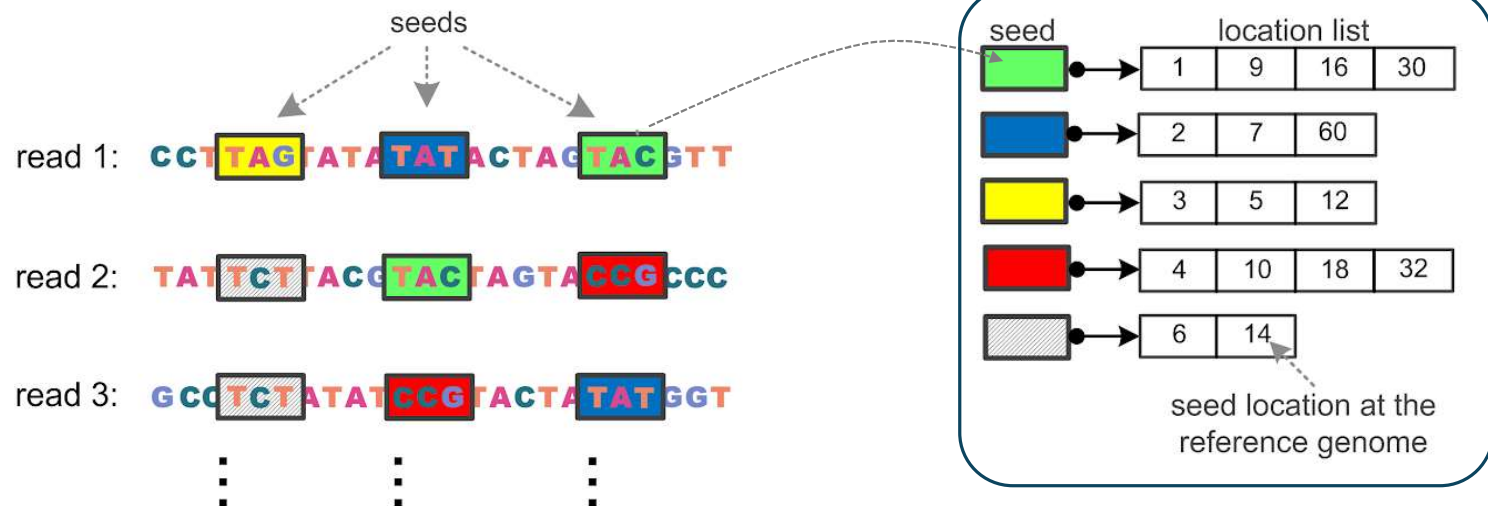
.FASTQ file:

```
@HWI-EAS209_0006_FC706VJ:5:58:5894:21141#ATCACG/1
T[redacted]AATAAATCT[redacted]TTAGATN[redacted]NNNNNNNNTAG
+
efcfffffcfeefffcfffffdddf`feed]`_]_Ba_^__[YBBBBBBBBBBRTT
```

Indeksiranje uz pomoć heširanja



Upiti korišćenjem indeksa



Poravnanje sekvenci

- Poklapanje se definiše kao minimalni broj promena koje su potrebne da bi se od ulaznog niza dobio referentni

organization x operation

Ref	o	-	-	r	g	a	n	i	z	a	t	i	o	n
Read	o	p	e	r	-	-	-	-	-	a	t	i	o	n

Ref	o	-	-	r	g	a	n	i	z	a	t	i	o	n
Read	o	p	e	r	-	a	-	-	-	-	t	i	o	n

Edit distance = 7

match
deletion
insertion
mismatch

organization x translation

Ref	o	r	g	a	n	i	z	-	a	t	i	o	n
Read	t	r	-	a	n	-	s	l	a	t	i	o	n

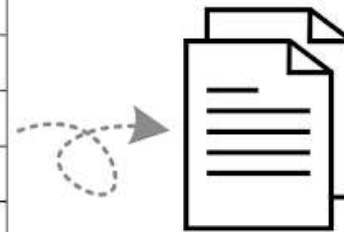
Ref	o	r	g	a	n	-	i	z	a	t	i	o	n
Read	t	r	-	a	n	s	l	-	a	t	i	o	n

Ref	o	r	g	a	n	i	z	a	t	i	o	n
Read	t	r	-	a	n	s	l	a	t	i	o	n

Edit distance = 4

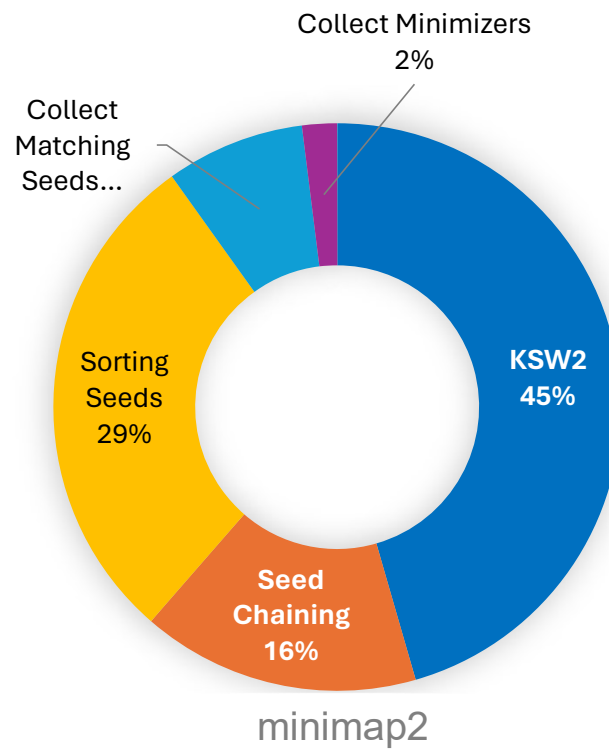
Dinamičko programiranje

		C	G	T	T	A	G	T	C	T	A	...
		0	0	0	0	0	0	0	0	0	0	
C		0	2	2	2	2	2	2	2	2	2	
C		0	2	3	3	3	3	3	3	4	4	
T		0	2	3	5	5	5	5	5	5	6	
T		0	2	3	5	7	7	7	7	7	7	
A		0	3	3	5	7	9	9	9	9	9	
G		0	2	4	5	7	9	11	11	11	11	
T		0	2	4	6	7	9	11	13	13	13	
A		0	2	4	6	7	9	11	13	14	14	15
T		0	2	4	6	8	9	11	13	14	16	16
	⋮											



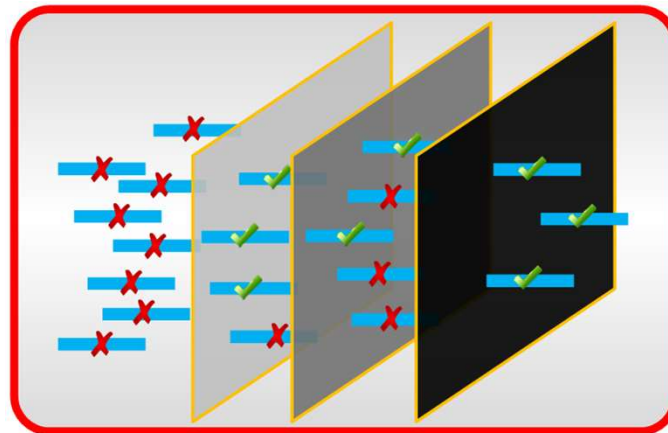
.bam/.sam file contains
necessary alignment
information (e.g., type,
location, and number of
each edit)

Izvršavanje mapiranja



Filtriranje nepotrebnih poredjenja

- Poravnanje sekvenci je skupa operacija
- Ideja: filtrirati stringove koji nisu slični
 - Filtrirati većinu stringova
 - Zadržati sva korektna mapiranja
 - Uraditi to efikasno



GateKeeper [2]

- Ako se dva stringa razlikuju u E promena, tada svaki bazni par može biti poravnat u najviše $2E$ pomeraja
- Koristi Hamingovu distancu
- Algoritam lak za implementaciju u hardveru
 - Generiše $2E + 1$ bitskih maski
 - Pretvori retke nule u 1 (101 u 111 i 1001 u 1111)
 - Uradi bitsko I nad svim maskama

Algoritam

Query :GAGAGAGATATTTAGTGTTCGAGCACTACAACACAAAAGAGGACCAACTTACGTGTCTAAAAGGGGGAACATTGTTGGGCCGGA
Reference :GAGAGAGATAGTTAGTGTTCGAGCCACTACAACACAAAAGAGGACCAACTTACGTGTCTAAAAGGGGAGACATTGTTGGGCCGG

Hamming Mask :	0000000000100000000000	111111011110001110110101101111111110001000	01	11011010010101
1-Deletion Mask :	111111111110011111011111	000	11	0000000000000000
2-Deletion Mask :	0000000010110111001111111111101111000111011010110111111111000100		10	11101101001010
3-Deletion Mask :	1111111111101110110011011101101101100010010011111111111100101100110		10	10111011101111
1-Insertion Mask :	11111111111011111011111101110110001001001111111111110010110011000		01	11101110111110
2-Insertion Mask :	0000001001111100111111110010001101010100110101111111111110111001		11	11000111101100
3-Insertion Mask :	1111111011101100110001111111110101101111110011001011101111111011		01	11010111001000

AND Mask : 00000000001000000000000100

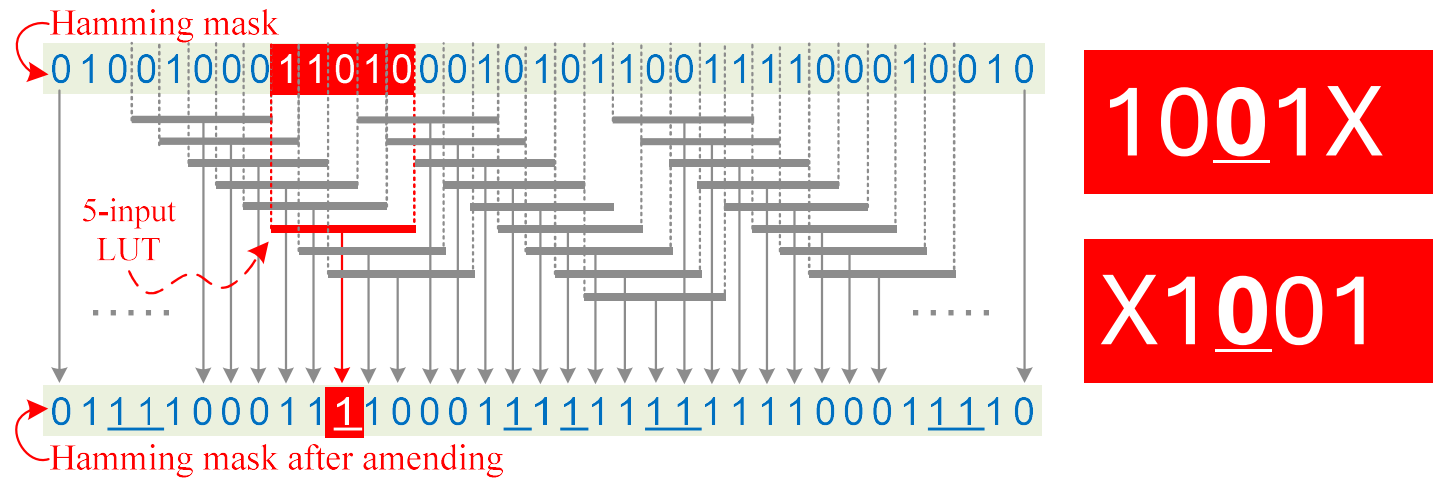
```
Hamming Mask :000000000100000000000111111111110001111111011111111111000100000111111111111111
1-Deletion Mask :11111111111111111111111100000000000000000000000000000000000000000110000000000000000
2-Deletion Mask :0000000011111111111111111111111100011111111111111111000100011111111111111110
3-Deletion Mask :111111111111111111111111111111110001111111111111111111111111111111111111111111
1-Insertion Mask :111111111111111111111111111111110001111111111111111111111111111110001111111111111110
2-Insertion Mask :000000111111111111111111111111110001111111111111111111111111111111111111111100011111100
3-Insertion Mask :11111111111111111111111111111111000111111111111111111111111111111111111111111111111000
```

[illegible]

Needleman-Wunsch Alignment : 

Implementacija

- Trivijalna, osim detekcije retkih nula



Performanse

Read length/E	mrFAST version/pre-alignment type	Filtering & verification time (speed-up)	Overall mapping time (speed-up)
100 bp /5 edits	2.1/No Pre-alignment	22.60 h (1×)	24.27 h (1×)
	2.6/Adjacency Filter	5.65 h (4×)	7.31 h (3.3×)
	2.1/GateKeeper	0.55 h (41×)	2.50 h (9.7×)
300 bp /15 edits	2.1/No Pre-alignment	0.94 h (1×)	1.02 h (1×)
	2.6/Adjacency Filter	0.04 h (24×)	0.12 h (8×)
	2.1/GateKeeper	0.003 h (279×)	0.09 h (11×)

Shouji [3]



- Tačno poravnanje uvek uključuje duge identične sekvence
- Ideja: koristiti pokretne preklapajuće prozore
 - Pronaći duge segmente uzastopnih nula
- Algoritam
 - Napraviti matricu suseda
 - Koristiti samo X suseda, gde je X mali broj
 - Pronaći najdužu dijagonalu nula i nju upisati u konačni rezultat

Algoritam (1)

<i>j</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>i</i>	G	G	T	G	C	A	G	A	G	C	T	C
1	G	0.	0.	1.	0.							
2	G	0.	0	1.	0.							
3	T	1.	1.	0	1	1						
4	G	0.	0	1.	0	1	0					
5	A		1.	1.	1	0	1	0				
6	G			1.	0	1	0	1	0			
7	A				1	0	1	0	1	1		
8	G				1	1	0	1	0	1	1	
9	T					1	1	1	1	1	0	1
10	T						1	1	1	1	0	1
11	G							1	0	1	1	1
12	T								1	1	0	1

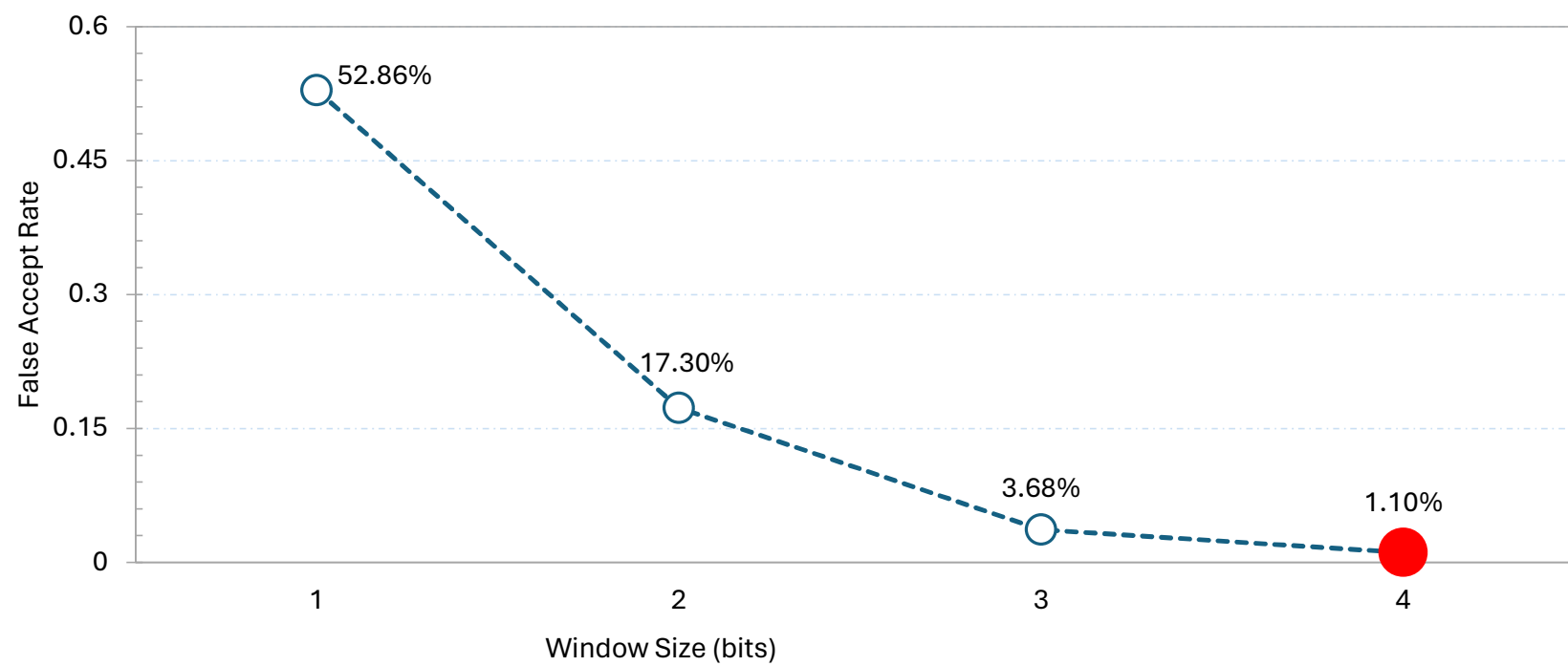
0	0	0	0	1	0	0	0	0	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---

Algoritam (2)

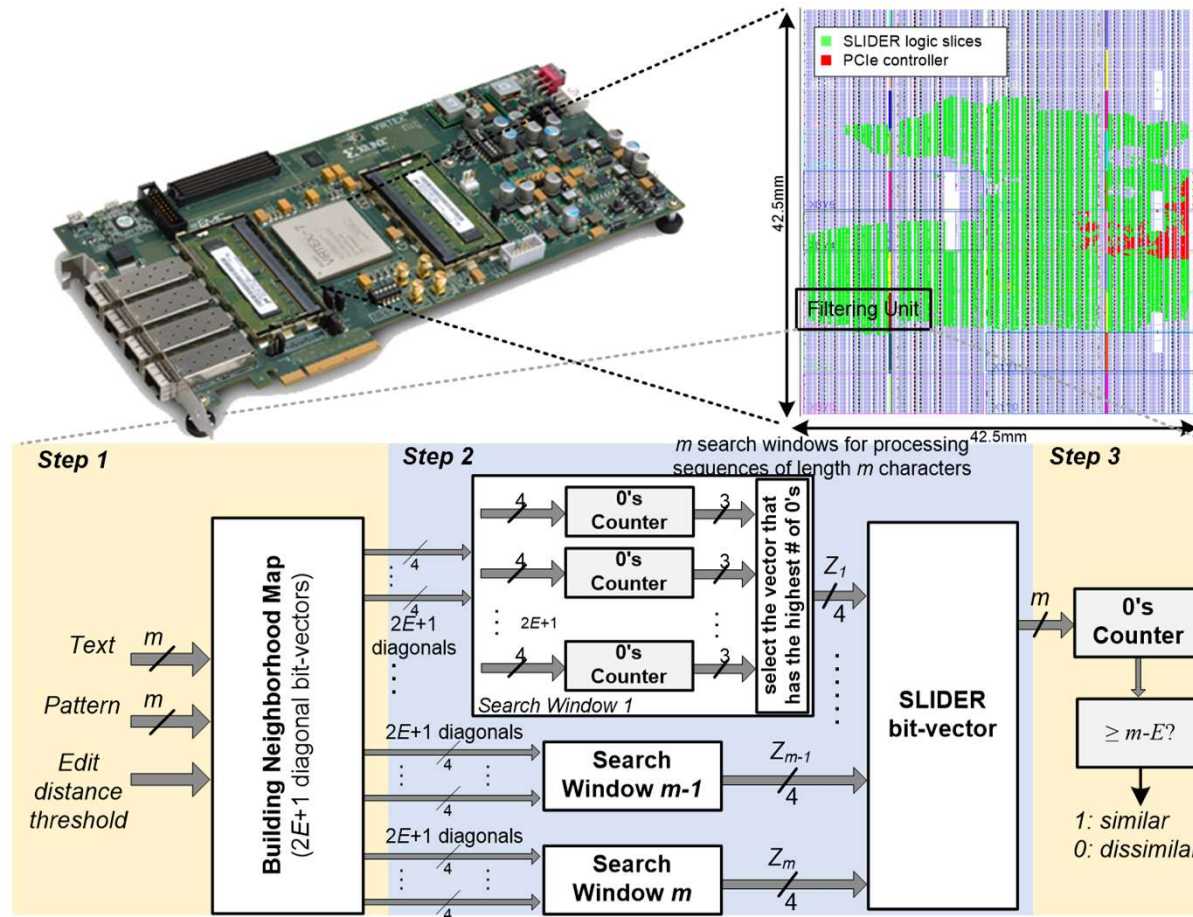
<i>j</i>	1	2	3	4	5	6	7	8	9	10	11	12
<i>i</i>	G	G	T	G	C	A	G	A	G	C	T	C
1	G	0	0	1	0							
2	G	0	0	1	0	1						
3	T	1	1	0	1	1						
4	G	0	0	1	0	1	0					
5	A		1	1	1	1	1	0				
6	G			1	0	1	0	1	0			
7	A				1	1	0	1	1	1		
8	G					1	0	1	0	1	1	
9	T						1	1	1	1	0	1
10	T							1	1	1	0	1
11	G								1	0	1	1
12	T									1	1	0

0	0	0	0	1	0	0	0	0	0	1	0	1
---	---	---	---	---	---	---	---	---	---	---	---	---

Veličina prozora



Implementacija



GenASM[4]

- Ubrzava računanje distanci
- Bazira se na Bitap algoritmu

Text	A	C	G	T	-
Exact Match	1111	1111	1111	1111	1111
1 Edit	0110	1010	1100	1110	1110
2 Edits	0000	0000	1000	1100	1100
3 Edits	0000	0000	0000	1000	1000
4 Edits	0000	0000	0000	0000	0000

For each character of the text (char):

Copy previous R bitvectors as oldR

$R[0] = (\text{oldR}[0] \ll 1) \mid \text{PM}[\text{char}]$

For $d = 1 \dots k$:

deletion = $\text{oldR}[d-1]$

substitution = $\text{oldR}[d-1] \ll 1$

insertion = $R[d-1] \ll 1$

match = $(\text{oldR}[d] \ll 1) \mid \text{PM}[\text{char}]$

$R[d] = \text{deletion} \& \text{mismatch} \& \text{insertion} \& \text{match}$

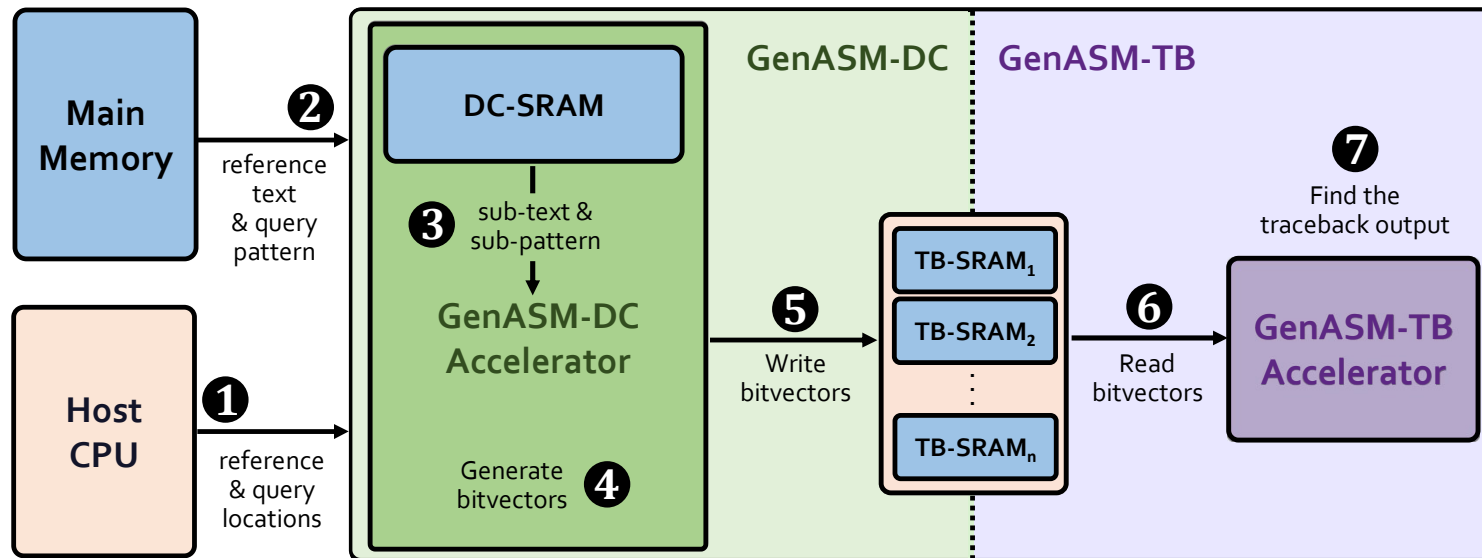
Check MSB of $R[d]$:

If 1, no match.

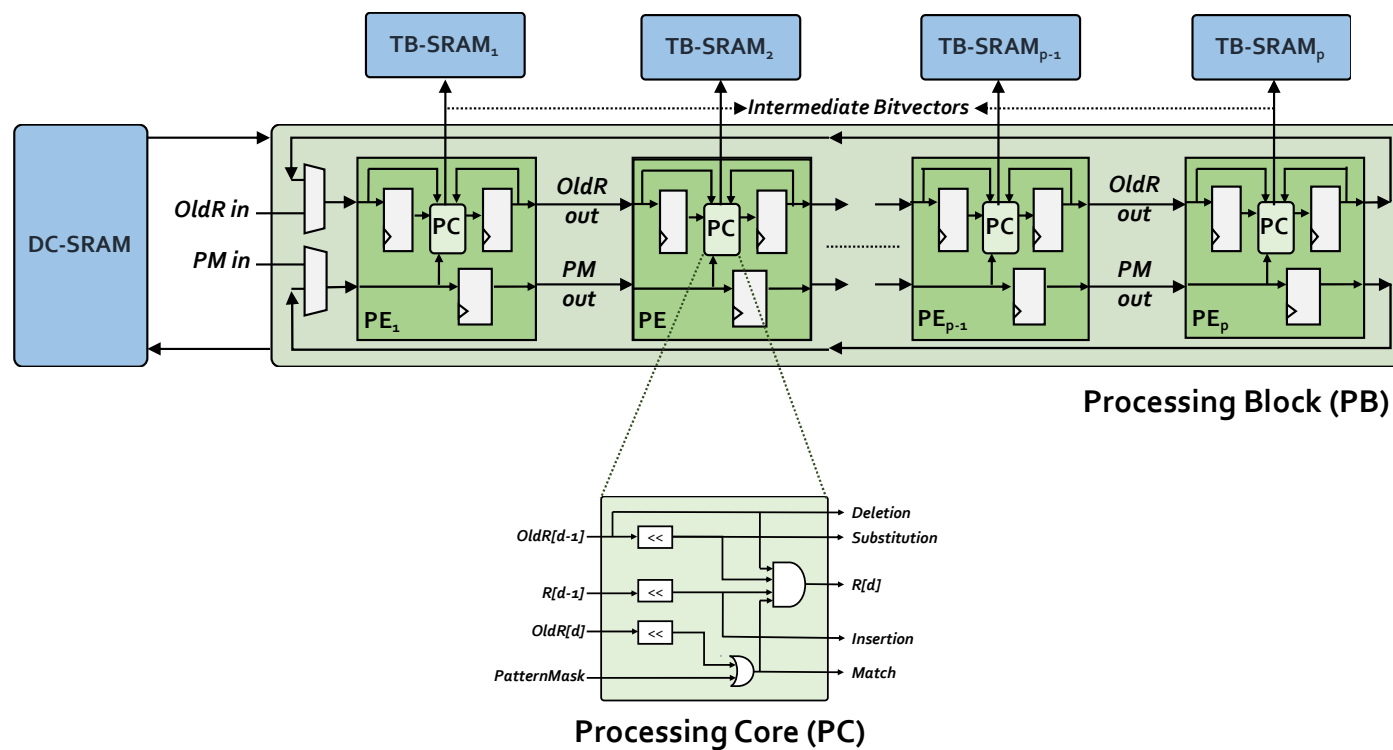
If 0, match with d many errors.

	A	C	G	T	
A	0	1	2	3	4
C	1	0	1	2	3
G	2	1	0	1	2
T	3	2	1	0	1
A	4	3	2	1	0

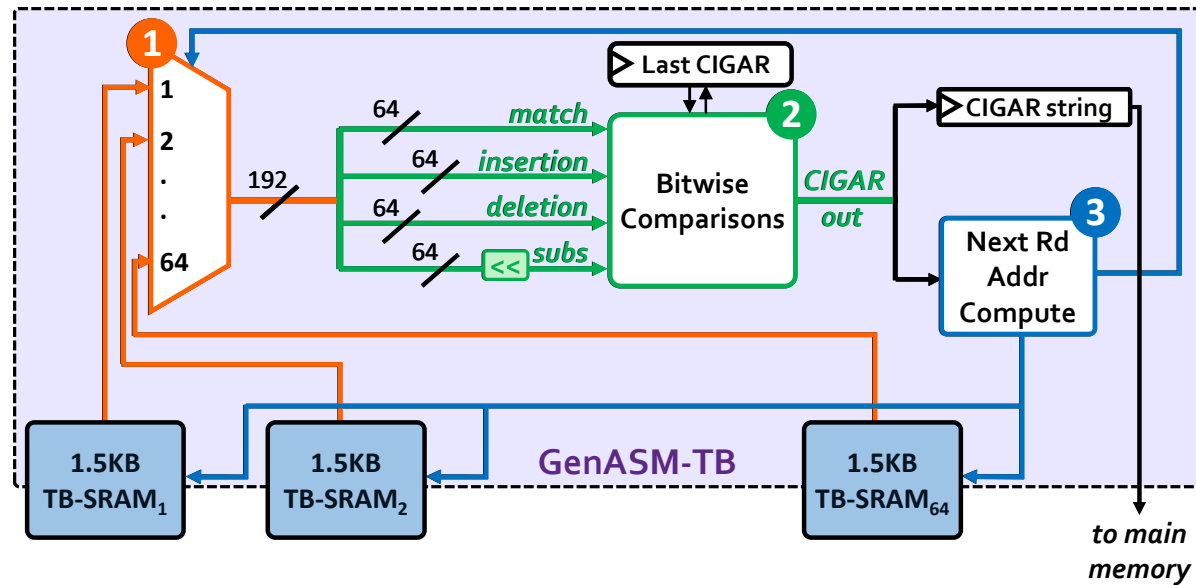
Arhitektura



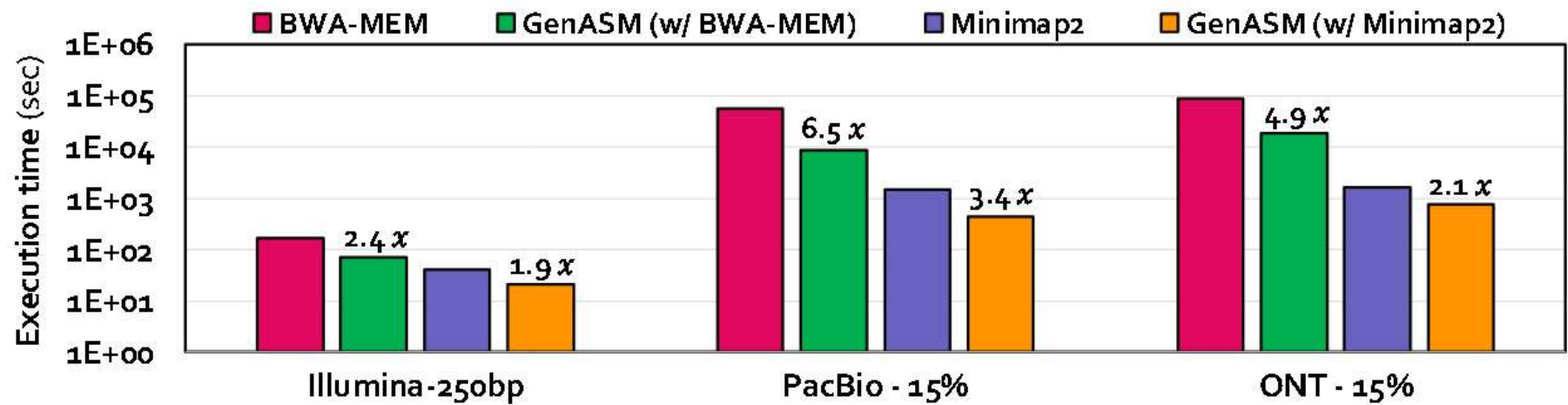
Implementacija računanja



Implementacija backtrace-a



Performanse



Reference

- [1] Mohammed Alser et al., "From molecules to genomic variations: Accelerating genome analysis via intelligent algorithms and architectures, " Computational and Structural Biotechnology Journal, 2022
- [2] Mohammed Alser et al., "GateKeeper: a new hardware architecture for accelerating pre-alignment in DNA short read mapping, " Bioinformatics, 2017
- [3] Mohammed Alser et al., "Shouji: a fast and efficient pre-alignment filter for sequence alignment," Bioinformatics, 2019
- [4] D. S. Cali et al., "GenASM: A High-Performance, Low-Power Approximate String Matching Acceleration Framework for Genome Sequence Analysis," MICRO, 2020